

TEAM: WATER NINJAS

Smart Water Level Monitoring For Industrial Tanks

Team Members:

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The Problem & Our Objective

The Problem:

- Millions of gallons wasted annually due to tank overflow
- Manual monitoring is unreliable and time-consuming
- No real-time awareness leads to costly waste

Our Objective:

- Enable real-time, continuous water level monitoring
- Trigger automatic overflow alerts before spillage occurs
- Reduce dependency on manual inspection entirely

Proposed Solution

A compact, low-cost embedded system with four key components:



Ultrasonic Sensor

Measures precise water level distance inside the tank without contact



Arduino UNO

Processes sensor data and calculates water level as a percentage



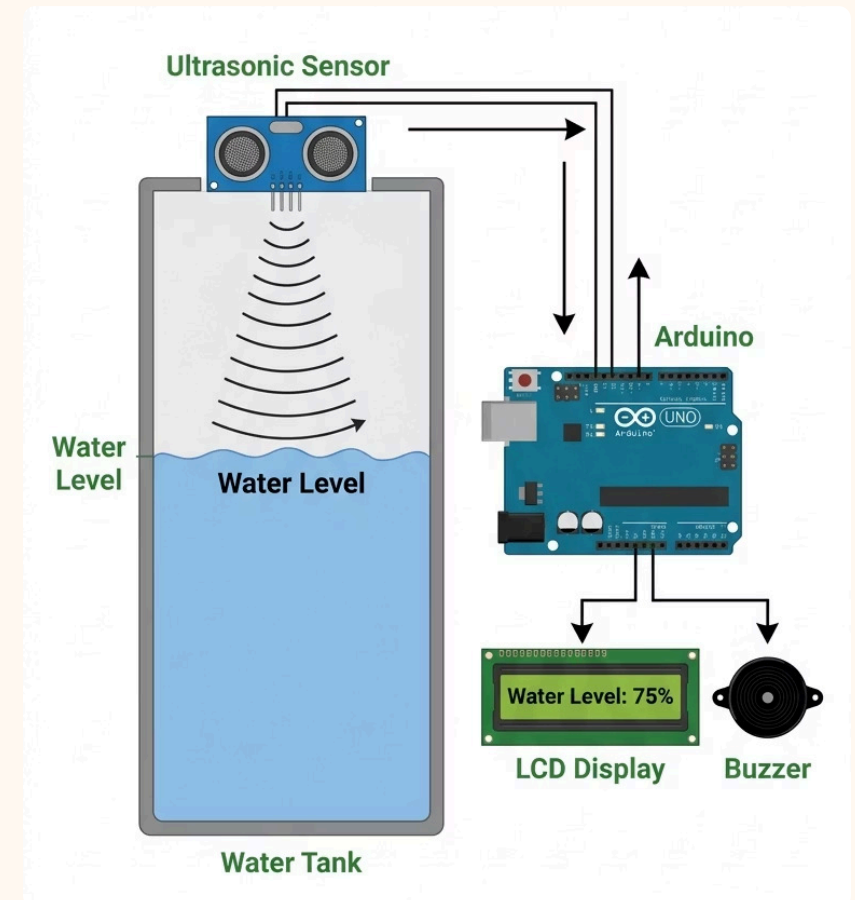
LCD Display

Shows live water level percentage for instant visual feedback



Buzzer Alert

Activates an audible alarm when water level reaches overflow threshold



Key Advantages & Applications

Key Advantages:

- Water-saving: Prevents overflow before it occurs
- Cost-effective: Built with widely available, affordable components
- Reliable & easy to install: Minimal wiring, no plumbing changes required

Where It's Used:

- 🏠 Homes - Rooftop and underground tank monitoring
- 🏢 Buildings - Multi-floor water supply management
- 🏭 Small Industries - Process water and coolant level control

Future Scope & Conclusion

What's Next:

- IoT Integration - Monitor water levels remotely via smartphone or web dashboard
- Automatic Motor Control - Auto-start and auto-stop pump based on water level thresholds
- Data Logging - Track usage trends for predictive maintenance and conservation

Conclusion

The Smart Water Tank Monitoring System delivers an efficient, reliable, and scalable solution to one of the most overlooked sources of daily water waste.

By combining embedded systems with real-time alerting, this project demonstrates how simple engineering can drive meaningful environmental and economic impact.

Built with Arduino · Ultrasonic Sensor · LCD · Buzzer